HK IMO Training 2019-2020
Problem Set 6
For 30 November 2019

Problem 21. Let ABC be an acute triangle, where $AB < AC$, and let M be the midpoint of BC, and K be the midpoint of the polygonal line BAC. Show that $\sqrt{2}KM > AB$.

Problem 22. The points D and E are chosen on the sides AB and AC of a triangle ABC so that $DB = BC = CE$. The segments BE and CD meet at P. Show that the circumcircles of the triangles BDP and CEP meet at the incentre of $\triangle ABC$.

Problem 23. In a class, there are m students. During September, each of them visited the pool several times. No one went to the pool twice in a day. It turned out that during September, all students visited the pool different number of times. Moreover, for any two students A and B there was a day when A visited the pool, while B did not, and vice versa. Determine the largest possible value of m.

Problem 24. Prove that for each positive integer n there are at most two pairs (a, b) of positive integers with following two properties:

(i) $a^2 + b = n$,

(ii) $a + b$ is a power of two, i.e. there is an integer $k \geqslant 0$ such that $a + b = 2^k$.